Actionable docker

Using docker to run packages locally.

Docker let's you do a lot of things, here's my tutorial on the same -

This tutorial is on actionable docker to start packages locally.

Installing Docker

Docker GUI is the easiest way to get off the ground.

You can find instructions to install docker on https://docs.docker.com/engine/install/

At the end of the installation, you need to make sure you're able to run the following command -

What are we using docker for?

Docker let's you do a lot of things.

It let's you containerise your applications.

It let's you run other people's code + packages in your machine.

It let's you run common software packages inside a container (For eg - Mongo, Postgres etc)

Where can we get packages from?

Just like you can push your code to Github/Gitlab.

You can push images to docker registries

Common commands to know

1. docker run
2. docker ps
3. docker kill

Running an image

1. Running a simple image

Let's say you wan't to run MongoDB locally
https://hub.docker.com/_/mongo

```
docker run mongo
```

You will notice you can't open it in MongoDB Compass .

Adding a port mapping

The reason is that you haven't added a port mapping

```
docker run -p 27017:27017 mongo
```

Starting in detached mode

Adding -d will ensure it starts in the background

```
docker run -d -p 27017:27017 mongo
```

Inspecting a container

```
docker ps
```

This will show you all the containers you are running.

Stopping a container

```
docker kill <container_id>
```

Will stop the container that you are running

Common packages

Mongo

```
docker run -d -p 27017:27017 mongo
```

Postgres

```
docker run -e POSTGRES_PASSWORD=mysecretpassword -d -p 5432:5432 pos
```

The connection string for this postgres would be

```
postgresql://postgres:mysecretpassword@localhost:5432/postgres
```

▼ Code to test it out

```
// Import the pg library
const { Client } = require('pg');

// Define your connection string (replace placeholders with your actual dat
const connectionString = 'postgresql://postgres:mysecretpassword@localh
```

ne connection string

```
const client = new Client({
                ectionString
});

// Connect to the database
client.connect(err => {
  if (err) {
    console.error('connection error', err.stack);
  } else {
    console.log('connected to the database');
  }
});

// Run a simple query (Example: Fetching the current date and time from Po
client.query('SELECT NOW()', (err, res) => {
  if (err) {
    console.error(err);
  } else {
    console.log(res.rows[0]);
  }

  // Close the connection
  client.end();
});
```